

Thromboangiitis Obliterans (Buerger's Disease): A Young Woman's Rare Case and Current Knowledge

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Abstract

In this paper, we described the clinical features and the typical vascular imaging picture of a rare case of thromboangiitis obliterans (TAO) in a 47-year-old, young, Caucasian woman without relevant medical records except for smoking habits and admitted to our Department of Internal Medicine with a necrotic lesion of the first and third toes of the right foot. After a rapid clinical and instrumental evaluation, necrotic tissue debridement was performed and the patient was treated with iloprost infusion and antibiotics (for overlapping infection) with clinical improvement. Moreover, we summarized the current knowledge about the disease making an update and critical examination of the literature. TAO, also known as Buerger's disease, is a rare disease. In recent decades, the incidence of TAO has decreased in Europe and the United States in men but several studies have reported an increase in the prevalence of disease in women. TAO is a segmental nonatherosclerotic inflammatory disorder that involves primarily small and medium arteries, veins, and nerves of the extremities with limb ischemia, pain, claudication, ulcers, and necrosis that can lead to amputation. Smoking is considered the main risk factor, but despite that, more than 100 years have passed since the first description of the disease by Leo Buerger, and knowledge on pathogenesis, precipitating factors, and therapy remain still unclear.

Keywords

- thromboangiitis obliterans
- Buerger's disease
- peripheral artery disease
- young woman

Thromboangiitis obliterans (TAO), also known as Buerger's disease, is a segmental nonatherosclerotic inflammatory disorder that involves primarily the small and medium arteries, veins, and nerves of the extremities. Smoking is considered the main precipitating factor, and this disease is more common in patients aged 35 to 50 years. von Winiwarter¹ provided the first description in 1878, but it was Leo Buerger who described the disease in detail in his article "Thromboangiitis Obliterans: A Study of the Vascular Lesions

Leading to Pre-Senile Spontaneous Gangrene" published in 1908 in *The American Journal of Medical Science*.²

Annual incidence of TAO is around 12% in the United States, but higher prevalence is found in the Middle East and the Far East regions. The prevalence of the disease among all patients with peripheral artery disease (PAD) varies from as low as 0.5 to 5.6% in Western Europe to as high as 45 to 63% in India, 16 to 66% in Korea and Japan, and 80% in Israel among Jews of Ashkenazi ancestry.³ The typical patient with TAO is a

man with a history of tobacco use, usually under 45 years of age, but in recent decades the incidence of the disease has decreased in men, possibly due to decreasing smoking population; in contrast, an increase in the incidence of TAO in women has been noted, between the years 1964 and 1970, the rate of TAO in women in one series was 5%, between 1970 and 1995 it rose to 9.3%, and remained relatively stable at 9.8% from 1995 to 2001, but recently several studies have reported an increase in the prevalence of the disease in women ranging from 11 to 23%.⁴

TAO is characterized by nonsuppurative vascular involvement with limb ischemia, pain, claudication, and thrombophlebitis which can cause ulcers, necrosis, and can lead to amputation. Amputation risk was estimated at 25% after 5 years past diagnosis, 38% after 10 years, and 46% after 20 years.⁵⁻¹⁷

In conclusion, TAO, also known as Buerger's disease, is a rare disease in Western countries. Recent observations have reported a decrease in the prevalence of the disease in men, but an increase in women, probably linked to an increase in smoking in young women; however lacking actually detailed knowledge on pathogenesis and risk factors of this disease, its prevalence is probably underestimated and TAO is classified in a simplistic way as PAD. Moreover, effective therapy, lacking randomized clinical trials dedicated to this medical condition, remains unclear.

Case Presentation

A 47-year-old Caucasian woman was admitted to our Department of Internal Medicine with a necrotic lesion of the first and third toe of the right foot (►Fig. 1). The patient reported in the months preceding admission recurrent ulcers to the right toes. She had no diabetes or other vascular

risk factors except an important cigarette smoking habit (two packs per day); she denied trauma. On physical examination, the tibial arteries of the right leg were hardly palpable. Based on these findings and considering that the patient was young with an important smoking history, TAO was suspected.

A vascular ultrasound examination of the arteries of the lower limbs was performed and it showed in the right lower limb normal femoral and popliteal arteries without focal accelerations of blood flow and triphasic Doppler signal, but tibial arteries were filiform with monophasic Doppler signal.

Computed tomography angiography (►Fig. 2) confirmed these results showing in right lower limb patency of the common, superficial and deep femoral arteries and of the popliteal artery; the anterior and posterior tibial arteries were pervious in the middle tract, but they were not recognizable in distal tract (probably occluded). The interosseous artery was filiform. In the left lower limb patency of the common, superficial and deep femoral arteries and of the popliteal artery have been described. The anterior tibial and interosseous arteries were filiform, and the posterior tibial artery was pervious in the middle tract, but it was not recognizable in its distal tract (probably occluded). Venous collateral circles were bilaterally described. Chest X-ray and standard electrocardiogram findings were normal. Laboratory findings were in the normal range or showed nonspecific results: C-reactive protein 10.4 mg/L (normal range <5), D-dimer 2,513 µg/L (normal range <500), and glycated hemoglobin 35 mmol/mol.

Immunologic Tests

Antinuclear antibodies, extractable nuclear antigens, lupus anticoagulant, anticardiolipin antibodies, and antibodies against β2-glycoprotein I were negative. **HLA typing was not performed.** At the level of the necrotic lesion of the toe,



Fig. 1 Images of the right foot with initial necrosis of the first and third toes.

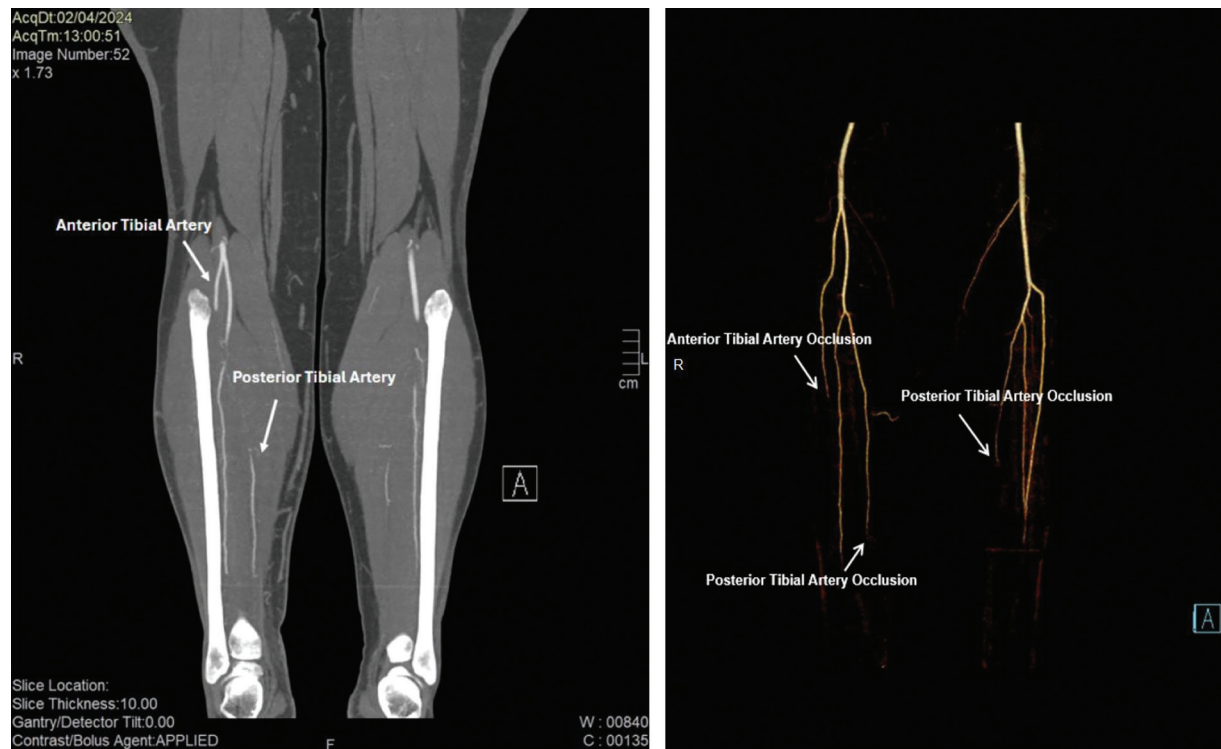


Fig. 2 CT angiography images showing the occlusion of the posterior and anterior right tibial arteries in the distal tract and the occlusion of the posterior left tibial artery in the mid-distal tract.

Staphylococcus aureus and *Enterobacter hormaechei* were detected. After necrotic tissue debridement, the patient was treated with antibiotics (Levofloxacin and Piperacillin-Tazobactam) and Iloprost infusion with clinical improvement. **The patient was convinced to quit smoking.** After 15 days of hospitalization, the patient was discharged and addressed to the Angiology Unit for follow-up.

Factors Contributing to Thromboangiitis Obliterans Pathogenesis

Smoking tobacco is central to the initiation, maintenance, and progression of TAO, but it may also develop as a result of chewing tobacco or marijuana use. Smoking cessation, but not decreasing the number of cigarettes per day can slow down disease progression.¹⁸ The possible relationships between tobacco use and TAO might include allergic-hyperergic reaction to antigens in cigarette smoke with inflammation activation, endothelial injury, and dysfunction; moreover, smoking tobacco can induce oxidative stress with an impact on many physiological and pathological activities closely related to TAO, such as inflammation, infection, and immunity, but actually the most effective targets of each signaling pathway remain in many aspects unknown and the exact pathogenesis of TAO and its association with smoking remains still unclear.^{19,20}

Studies showed that nearly two-thirds of patients with TAO had a periodontal infection.²¹ Chen et al²² described that periodontal abnormalities were more frequent in patients with TAO than in smokers free of this condition (89.5 vs. 28%). Iwai et al²³ detected seven kinds of bacterial

DNA in oral samples and occluded arteries suggesting an association between chronic oral infection and TAO. Kubota et al²⁴ observed in rats 83.3% of thrombosis in popliteal artery specimens 4 weeks after the intravenous administration of *Porphyromonas gingivalis*. There is a possibility that bacteria can induce the synthesis of autoantibodies as suggested by the higher prevalence of antibodies against β 2-glycoprotein I in patients with TAO and severe periodontal disease. The existence of several autoantibodies in some TAO patients has been reported, including anti-endothelial cell antibodies, anti-elastin, and anti-collagen I and III antibodies,^{25–28} but other authors argued that this association is not convincing.²⁹

Some studies suggest that susceptibility to TAO, as in other diseases, is in part, controlled by genes involved in innate and adaptive immunity.³⁰ Incidence of HLA A-9 and HLA B-5 seems to be higher in patients with TAO than in healthy persons, whereas HLA-B12 seems to be lower. Besides HLA typing, several studies on the single nucleotide polymorphism of nitric oxide synthase, endothelin 1, plasminogen activator inhibitor, platelet receptors, homocysteine, Factor V, prothrombin, Toll-like receptor 4, and CD14 have been conducted in TAO patients. However, only polymorphisms of Myeloid differentiation factor 88 and endothelial nitric oxide synthase seem to play a role in the development of the disease. Chen et al³¹ found that the frequencies of CD 14 TT genotype, HLA-DRB 1*1501, and HLA-DPB 1*0501 were significantly higher in patients with TAO than in healthy controls and Shi et al³² studying genetic factors potentially involved in the pathogenesis of the disease found that SNP rs376511 in interleukin 17 receptor C,

rs7632505 in semaphorin 5B, and rs10178082 correlated with a significant risk of TAO in the Uyghur population in China.

Clinical Findings and Diagnosis

TAO is a segmental inflammatory, thrombotic occlusive peripheral vascular disease that involves the medium and small-sized vessels in young smokers. Patients with TAO typically present ischemic symptoms. Involvement of both the upper and lower extremities and the size and location of affected vessels help to distinguish it from atherosclerosis. TAO often presents as intermittent claudication, rest pain, ulcerations, and digital gangrene occurs in more advanced stages of the disease. Raynaud's phenomenon is present in 40% of patients. Superficial thrombophlebitis and peripheral nerve involvement with sensory findings may be present. Sasaki et al³³ analyzed the clinical features of 850 patients with TAO. Clinical manifestations were Fontaine's classification I in 8.0%, Fontaine II in 29.6%, Fontaine III in 23.9%, and Fontaine IV in 38.1%. Ulcer formation in the history of patients had occurred in 45.2%, mainly in the toes (85.9%). In these patients, failure of smoking cessation significantly affected the risk of ulcer formation (odds ratio = 1.71, 95% CI = 1.19–2.47; $p = 0.004$) and amputation (odds ratio = 2.73, 95% CI = 1.86–4.01; $p < 0.0001$). Abstinence from tobacco significantly reduced the risk.

Ates et al³⁴ in 344 Turkish patients with TAO described foot coldness in 312 (90.6%) patients, color changes in 290 (84.3%), rest pain in 160 (46.5%), claudication in 166 (48.2%), and necrotic ulcers in 185 (53.1%).

There are no specific diagnostic criteria for TAO diagnosis accepted by the scientific community, even if several criteria have been proposed.^{5,35,36} Common features of these diagnostic criteria include detecting the disease at 40 to 50 years of age, tobacco smoking, and the presence of clinical, ultrasonographic, and angiographic characteristics along with the exclusion of diabetes or autoimmune disorders.

We believe that the following criteria for the diagnosis of TAO published by the Japanese Ministry of Health may be exhaustive.³³

1. *Clinical signs (at least one to two of the following signs are required)*
 - Drop in skin temperature, paresthesia, and Raynaud's syndrome involving the distal parts of the upper or lower limbs
 - Intermittent claudication
 - Resting pain in soles and palms
 - Painful ulceration or gangrene
 - Migrating superficial phlebitis
2. *Physical examination (at least one to two of the following findings are required)*
 - Lower temperature of the skin in the upper or lower limbs, or the distal parts of the toes or fingers
 - Impaired arterial pulsation in the peripheral segment of a vessel, with retained proximal pulse

- Lower value of ankle brachial index (ABI) determined by Doppler method.

3. *Laboratory tests: The results of routine laboratory tests are usually normal in TAO patients.*
4. *Arteriography (at least one to two of the following findings are required)*
 - Multiple segmental occlusions of distal arterial sections (distally from the elbow and knee)
 - Chronic vascular occlusion due to secondary thrombosis
 - An absence of atherosclerotic lesions, such as calcification of vascular walls
 - Tapering and abrupt obstruction of vessels, a twisting course of the involved vessels
 - "Corkscrew" or "tree-root" collaterals
5. *Absence of other vascular disorders (the following have to be excluded)*
 - Atherosclerosis obliterans
 - Traumatic arterial thrombosis
 - Popliteal entrapment syndrome
 - Vascular disorders
 - Behçet's disease

Therapy

The prognosis for patients depends largely on the ability to discontinue tobacco use. Cessation smoking results in dramatic improvement. Prostacyclin or its analogs (iloprost, beraprost, and reprostinil sodium) are used to manage TAO. Iloprost is considered the most effective.³⁷ A placebo-controlled, double-blind, study in 319 TAO patients with rest pain, trophic lesions, or both showed that iloprost was significantly more effective than placebo for relief of rest pain without the need for analgesics, but showed no significant effects versus placebo on total healing of lesions.³⁸ Bozkurt et al³⁹ compared the effects of iloprost and lumbar sympathectomy (LS) in the treatment of Buerger's disease. The complete healing rate was 61.9% in the iloprost group versus 41% in the LS group at the 4th week ($p = 0.012$), respective values for the 24th week were 85.3 and 52.3% ($p < 0.001$). The size of the ulcers decreased more in the iloprost group. Authors concluded that using iloprost relieves ischemic symptoms better than LS and that in the era of stable prostacyclin analogs, there is no reliable evidence to support the use of LS in Buerger's disease. However, it is important to emphasize that the usefulness of surgical sympathectomy seems to be dependent on smoking cessation and its role in TAO remains controversial because there are no long-term data to affirm its benefit.

De Haro et al⁴⁰ performed a prospective, open-label study with Bosentan, a dual Endothelin-1 receptor antagonist, in patients with TAO and ulcers refractory to conventional treatment concluding that Bosentan may result in a long-term improvement of clinical and endothelial function outcomes for treating ischemic ulcers in

Buerger's disease patients with an acceptable tolerability profile.

Autologous bone marrow-derived stem cell therapy (ABMSCT) is an emerging modality to induce angiogenesis from endothelial progenitors. Boda et al⁴¹ treated five patients with PAD (four with TAO) with ABMSCT using isolated CD34+ cells with characterized phenotype administered by intramuscular injections. Significant improvement in pain and walking distance was observed in all patients. Nonhealing ulcers disappeared in three patients and became smaller and thinner in one patient. ABI improved significantly on the treated limb but did not change on the contralateral limb. New collaterals were detected by angiography in three patients. Boda et al concluded that ABMSCT with isolated CD34+ cells is safe, effective, and results in sustained clinical benefit for patients with PAD. These results were confirmed by another study⁴² that retrospectively analyzed the data of 58 limbs from 37 patients (mean age, 43.0 years; range, 28–63 years; male, 91.9%) with Buerger's disease who were treated with ABMSCT. At 6 months, patients demonstrated significant improvements in the Rutherford category, pain score, pain-free walking time, and toe brachial pressure index. A total of 76.5% of ischemic wounds achieved complete or improved healing. Administration of growth factors to patients with TAO increases the concentration of angiogenic factors in the lower ischemic limb, aiming to improve endothelial cell proliferation, migration, and blood vessel formation, and could represent an effective future therapy.

Also, electrical stimulation of the posterior column and Spinal Cord Stimulation (SCS) may have a role in TAO management. Multiple articles have been published on SCS use in nonoperable critical limb ischemia and improved limb survival has been reported. Stimulation at T10–L1 level induces anesthesia in the lower extremity alleviating ischemic pain. Donas et al⁴³ studied 29 patients with TAO treated with SCS and showed a significant benefit in the microcirculation, a good limb survival rate, and the absence of new trophic lesions and concluded that because of the diffuse, distal, segmental nature of the disease, SCS should be considered an alternative treatment modality in patients with Buerger's disease. However, the data available about SCS use in TAO patients are few, and the benefits are unclear.

Surgical revascularization is usually not feasible in patients with TAO because of the distal and diffuse nature of the disease. Sometimes, bypass surgery may be considered in select patients with severe ischemia and suitable distal target vessels, but these patients often have suboptimal outcomes.⁴⁴ Graziani et al⁴⁵ performed in 17 TAO patients (20 limbs) an extended angioplasty of each tibial and foot artery obstruction to achieve direct perfusion of at least one foot artery.

Amputation-free survival with no need for major amputation and sustained clinical improvement was achieved in 16 limbs (84.2%). These results were confirmed also by Yuan et al⁴⁶ even if in a small number of patients.

Conflict of Interest

None declared.

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